

CE 518: Structural Reliability and Risk Assessment

Course Code	CE 518
Course Name	Structural Reliability and Risk Assessment
L-T-P-C	3-0-0-3
Prerequisites	CE 351 – Design of Reinforced Concrete Structures
Intended for	UG/PG elective
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Review of Probability and Statistics:** Basic definition of probability, Concept of random variables; Concepts of probability functions - PDF, CDF, and PMF; Types of probability distributions; Correlations, Conditional probability, Bayes theorem [8 hours]
- **Structural Reliability Analysis 1:** Basic concepts, Exact solution, Mean value first-order second-moment (MVFOSM), First-order second-moment (FOSM), First-order reliability method (FORM), Sensitivity analysis. [8 hours]
- **Probabilistic Simulations:** Monte Carlo simulations, Stratified sampling, Importance sampling. [5 hours]
- **Structural Reliability Analysis 2:** Component and System reliability, Second-Order Reliability Method (SORM), Time-varying reliability, Response surface method, Introduction to machine learning tools. [8 hours]
- **Reliability-based design codes:** Introduction, LRFD, Calibration of safety factors. [6 hours]
- **Advanced Topics and Applications:** Risk assessment of civil structure and infrastructure systems under service loads and natural hazards; Reliability-based Optimization Problems; Bayesian Inference Techniques; Application in structural health monitoring - Concept of Value of Information. [7 hours]

Text Books

1. Nowak, A.S. & Collins K.R., **Reliability of Structures**, 2nd Edition, CRC Press, 2012.
2. Ranganathan R, **Structural Reliability: Analysis and Design**, Jaico Publishing, 2006.

References

1. Melchers RE, **Structural Reliability Analysis and Prediction**, 2nd ed., Wiley, 1999.
2. Haldar, A & Mahadevan, S, **Probability, Reliability, and Statistical Methods in Engineering Design**, Wiley, 1999.
3. Wang, C., **Structural Reliability and Time-dependent Reliability**, Cham, Switzerland: Springer, 2021.
4. Ditlevsen, O., & Madsen, H. O., **Structural Reliability Methods** (Vol. 178), Wiley, 1996.
5. Chandrasekaran, S, **Reliability and Risk Assessment**, CRC Press, 2016
6. Thoft-Christensen, P., and Murotsu, Y., **Application of Structural Systems Reliability Theory**, Springer Verlag, 1986.